PAPER–II

1. Construction Technology, Equipment, Planning and Management :

1.1 Construction Technology :

Engineering Materials—Physical properties of construction materials with respect to their use in construction-Stones, Bricks and Tiles; Lime, Cement, different types of Mortars and Concrete.

Specific use of ferro cement, fibre reinforced C.C, High strength concrete.

Timber, properties and defects-common preservation treatments.

Use and selection of materials for specific use like Low Cost Housing, Mass Housing, High Rise Buildings.

1.2 Construction—Masonry principles using Brick, stone, Blocks–construction detailing and strength characteristics.

Types of plastering, pointing, flooring, roofing and construction features. Common repairs in buildings.

Principles of functional planning of building for residents and specific use-Building code provisions.

Basic principles of detailed and approximate estimating-specification writing and rate analysis – principles of valuation of real property.

Machinery for earthwork, concreting and their specific uses–Factors affecting selection of equipments–operating cost of Equipments.

1.3 Construction Planning and Management : Construction activity–schedules-organization for construction industry–Quality assurance principles.

Use of Basic principles of network–analysis in form of CPM and PERT –their use in construction monitoring, Cost optimization and resource allocation.

Basic principles of Economic analysis and methods.

Project profitability–Basic principles of Boot approach to financial planning–simple toll fixation criterions.

2. Surveying and Transportation Engineering :

2.1 Surveying—Common methods and instruments for distance and angle measurement for CE work–their use in plane table, traverse survey, leveling work, triangulation, contouring and topographical map.

Basic principles of photogrammetry and remote sensing.

2.2 Railway Engineering—Permanent way–components, types and their functions – Functions and Design constituents of turn and crossings–Necessity of geometric design of track–Design of station and yards.

2.3 Highway Engineering—Principles of Highway alignments–classification and geometrical design elements and standards for Roads.

Pavement structure for flexible and rigid pavements-Design principles and methodology of pavements.

Typical construction methods and standards of materials for stabilized soil, WBM, Bituminous works and CC roads.

Surface and sub-surface drainage arrangements for roads-culvert structures.

Pavement distresses and strengthening by overlays.

Traffic surveys and their applications in traffic planning-Typical design features for channelized, intersection, rotary etc–signal designs–standard Traffic signs and markings.

3. Hydrology, Water Resources and Engineering :

3.1 Hydrology—Hydrological cycle, precipitation, evaporation, transpiration, infiltration, overland flow, hydrograph, flood frequency analysis, flood routing through a reservoir, channel flow routing-Muskingam method.

3.2 Ground Water Flow—Specific yield, storage coefficient, coefficient of permeability, confined and unconfined equifers, aquifers, aquitards, radial flow into a well under confined and unconfined conditions.

3.3 Water Resources Engineering—Ground and surface water resource, single and multipurpose projects, storage capacity of reservoirs, reservoir losses, reservoir sedimentation.

3.4 Irrigation Engineering—(i) Water requirements of crops—consumptive use, duty and delta, irrigation methods and their efficiencies.

(ii) Canals—Distribution systems for canal irrigation, canal capacity, canal losses, alignment of main and distributory canals, most efficient section, lined canals, their design, regime theory, critical shear stress, bed load.

(iii) Water logging—Causes and control, salinity.

(iv) Canal structures—Design of, head regulators, canal falls, aqueducts, metering flumes and canal outlets.

(v) Diversion headwork—Principles and design of weirs of permeable and impermeable foundation, Khosla's theory, energy dissipation.

(vi) Storage works—Types of dams, design, principles of rigid gravity, stability analysis.

(vii) Spillways—Spillway types, energy dissipation.

(viii) River training—Objectives of river training, methods of river training.